

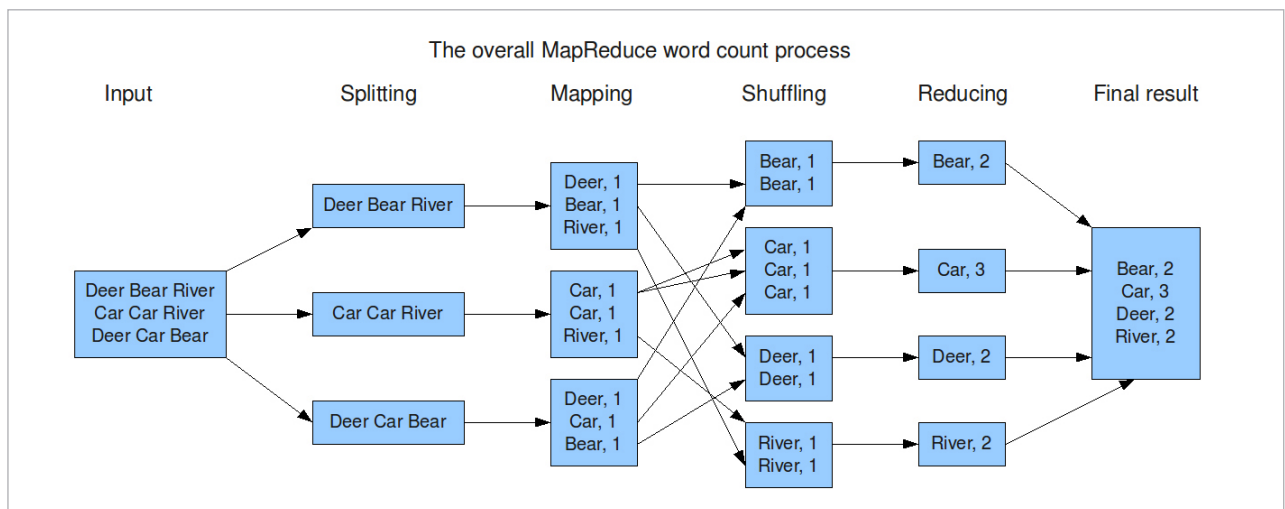


Figure 3: MapReduce method (Courtesy: Today Software Magazine)

MapReduce

Storing and accessing Big Data, with all its complications and difficulties, is still the easy part. The real fun begins when we try to process Big Data. This is where Hadoop does its magic with an algorithm called MapReduce. MapReduce is exactly what it sounds like. We map the data, followed by reducing it. This is a five step process, as shown in Figure 3.

MapReduce takes the data as input and splits it into different parts. It then maps each individual component within each part. In the shuffle phase, the data is shuffled and sorted into clusters of homogenous data. This data is then reduced by integrating homogenous data and assigning IDs, which give the smaller final output.

YARN

Short for 'Yet Another Resource Negotiator', YARN is exactly what it sounds like. A Hadoop environment is constrained to the amount of resources it has, namely, CPUs, memory units, storage space, etc. But since Hadoop is designed to be able to run multiple processes at the same time, a mechanism for allocating resources is necessary to make sure there are no deadlocks and every task is accomplished in its desired time period.

As shown in Figure 4, the physical resources are saved in what is referred to as containers. When a job requires the resources to run, the application manager sends a request to the node manager to allocate the container. The node manager then sends the resources to the resource manager, which assigns

the container to the client process.

Spark

While HDFS, MapReduce and YARN are the three major defining components of Hadoop, the Hadoop ecosystem has more tools to offer when it comes to Big Data processing. One such tool is Apache Spark. Created by the Apache Foundation back in 2009, Spark is one of the most actively developed Big Data processing tools in the open source community.

Spark is one of the only few tools in the industry that combines Big Data analytics tools with machine learning and AI. This enables Spark to do complex data transformations and, at the same time, run machine learning algorithms on the data to create trained models. Companies like Alibaba, Amazon, 4Quant and other big names deploy Apache Spark to process and analyse their data.

Spark is 100 times faster than MapReduce in processing data. Since the Spark computations are done in the memory itself, it has the ability to perform both batch processing as well as real-time processing of data. Spark is also comparatively lightweight as most of it is written in Scala. As shown in Figure 5, there are five major

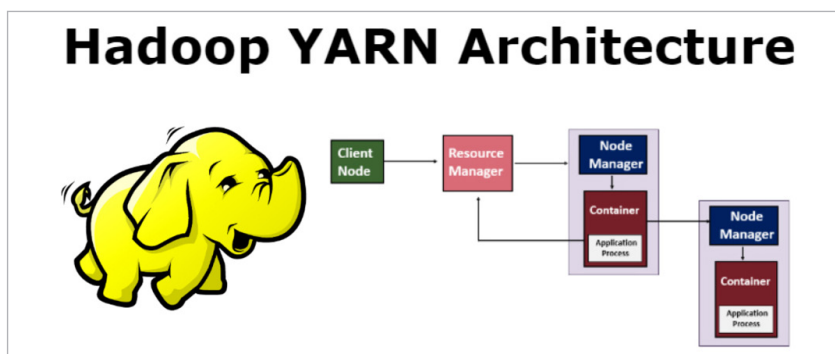


Figure 4: YARN architecture (Courtesy: eduCBA)